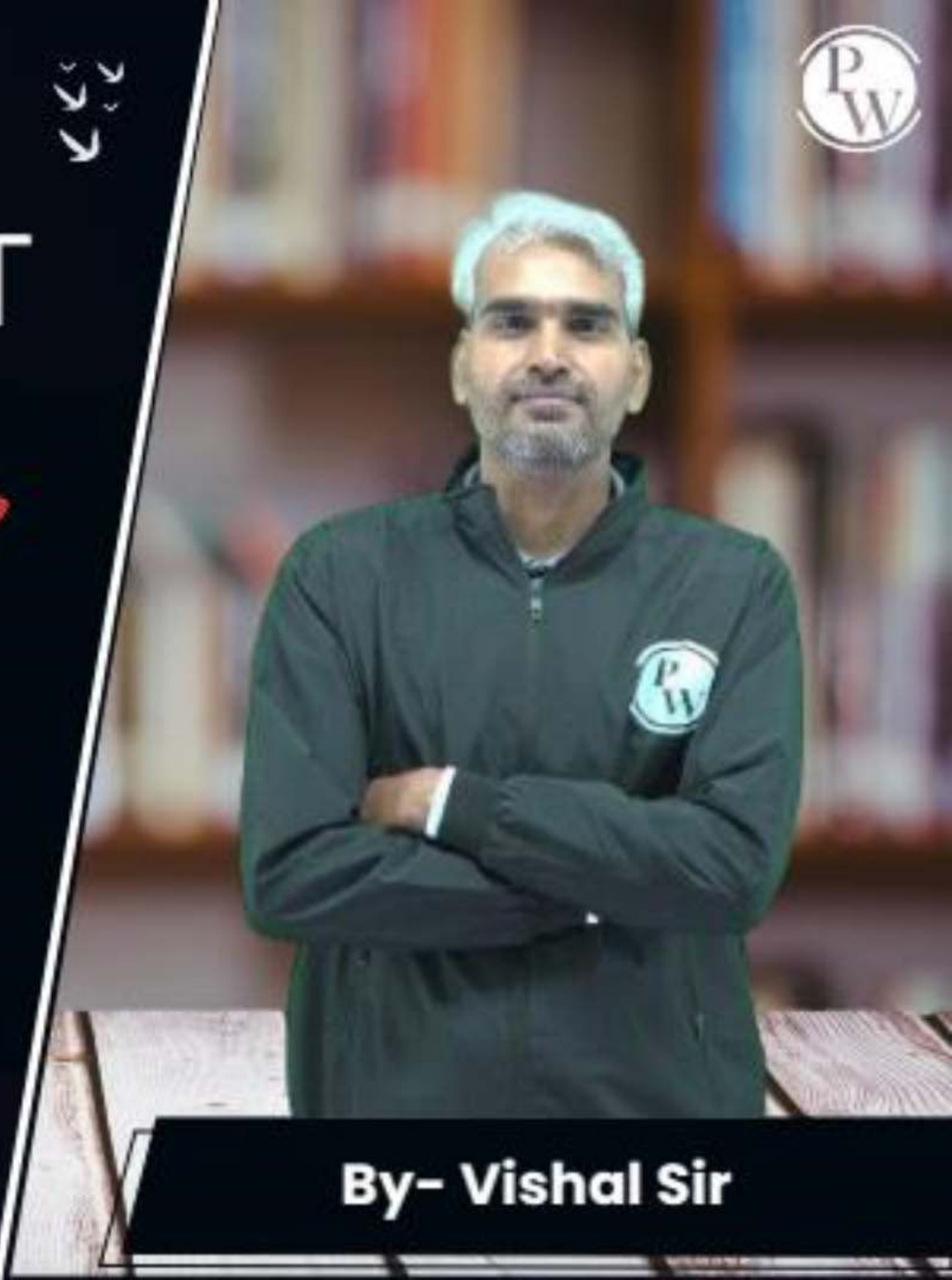
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 06



By- Vishal Sir

Recap of Previous Lecture



Topic — Self complementary graph

Topic — Graph isomorphism



Slide

Topics to be Covered

Topic	Planar graphs



Slide

Planar Graph



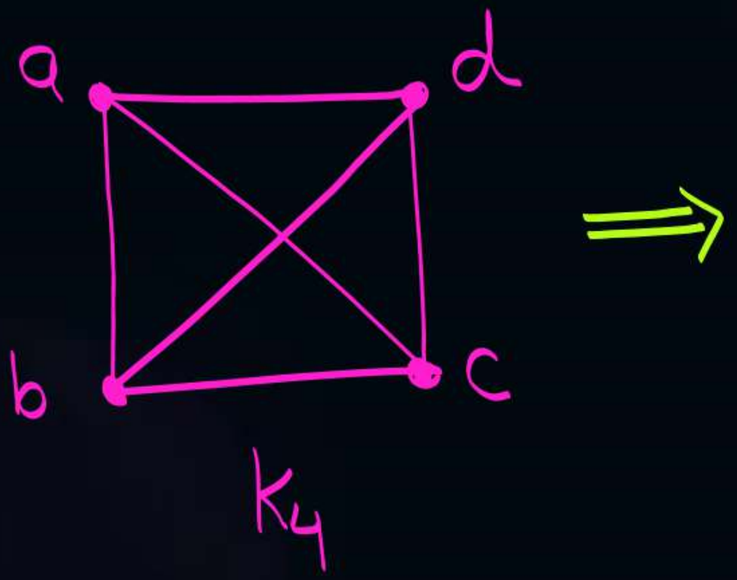
Topic : Planar graph

A graph G= (V, E) is said to be planar if it can be drawn in the plane so that no two edges of G intersect each other at a non-vertex point.

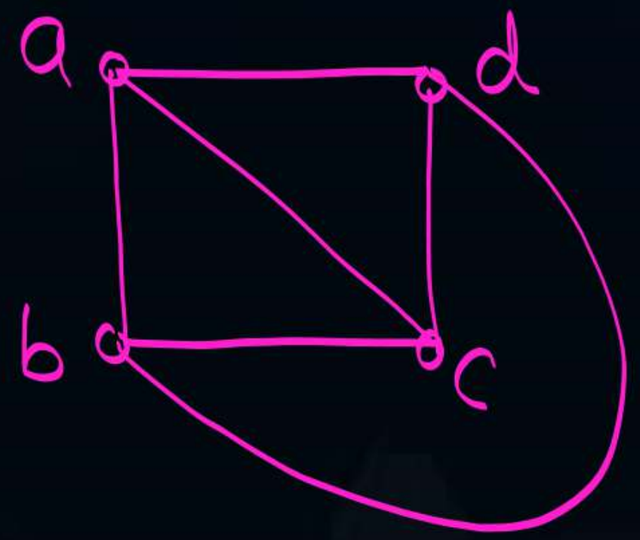
Such a drawing of a planar graph is called a planar embedding of the graph.



Topic : Planar graph



K_4

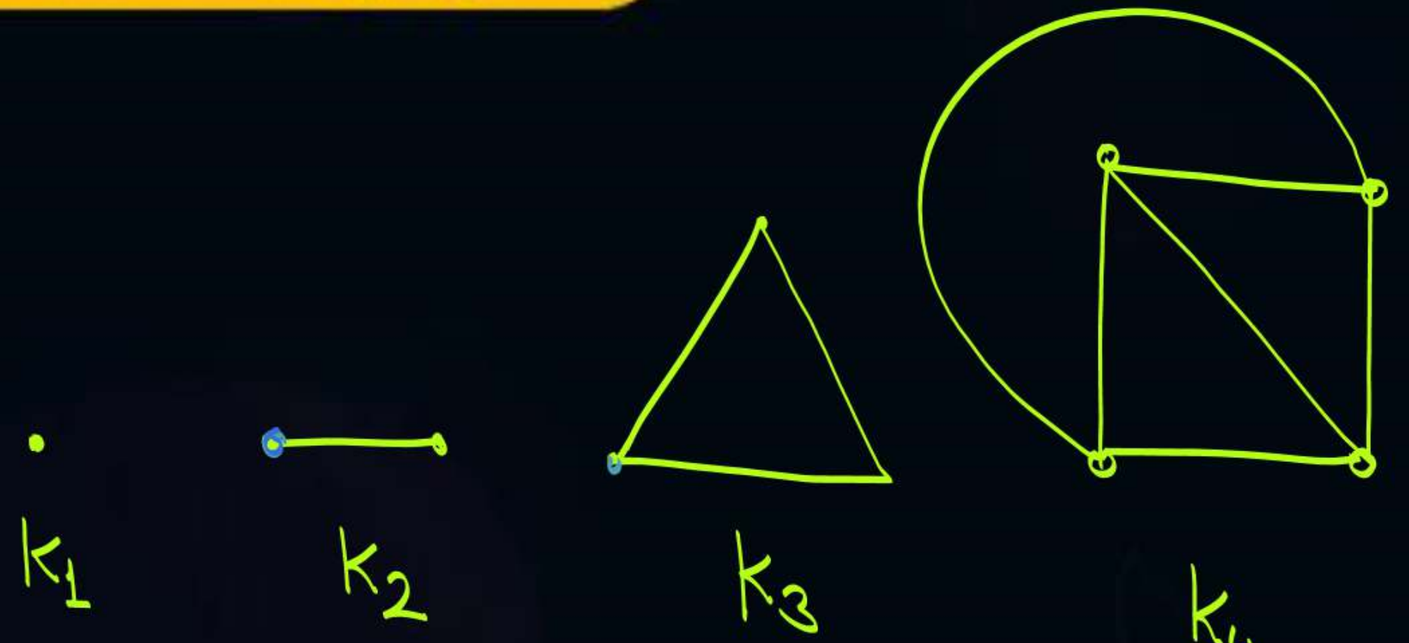


We can draw K_4 on a plane s.t. no two edges cross each other at a non-vertex point.

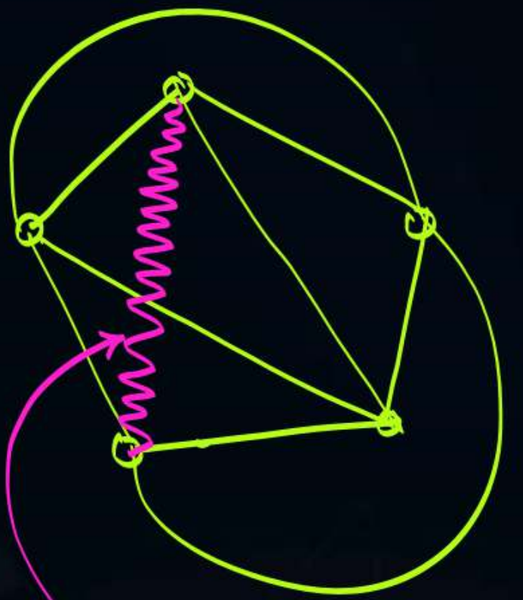
∴ K_4 is planar



Topic : Planar graph

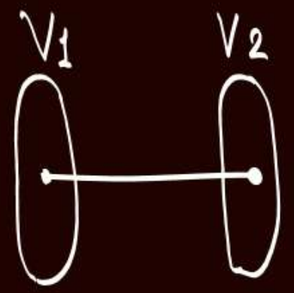


Complete graph K_n is planar if and only if $n \leq 4$

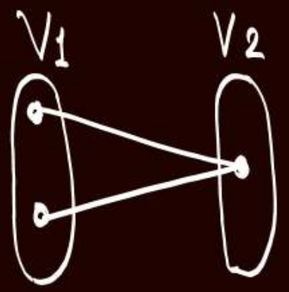


K_5 is not planar

- K_5 is a non-planar graph with '5' vertices.

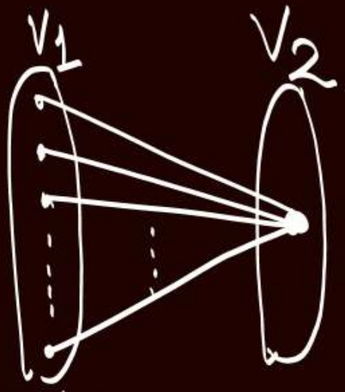


$K_{1,1}$ is planar



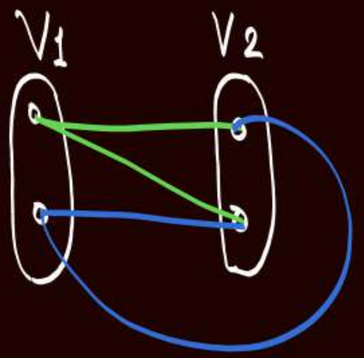
$K_{2,1}$ is planar

$\therefore K_{1,2}$ is also planar

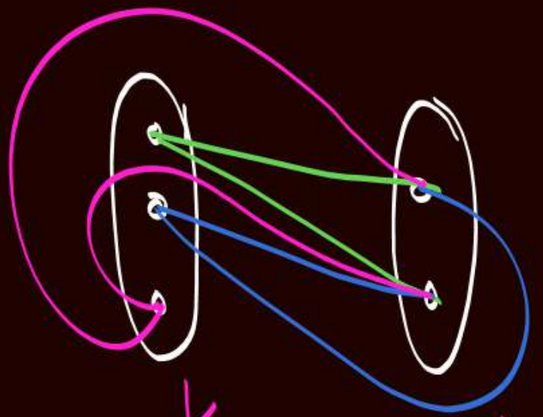


$K_{n,1}$ is planar

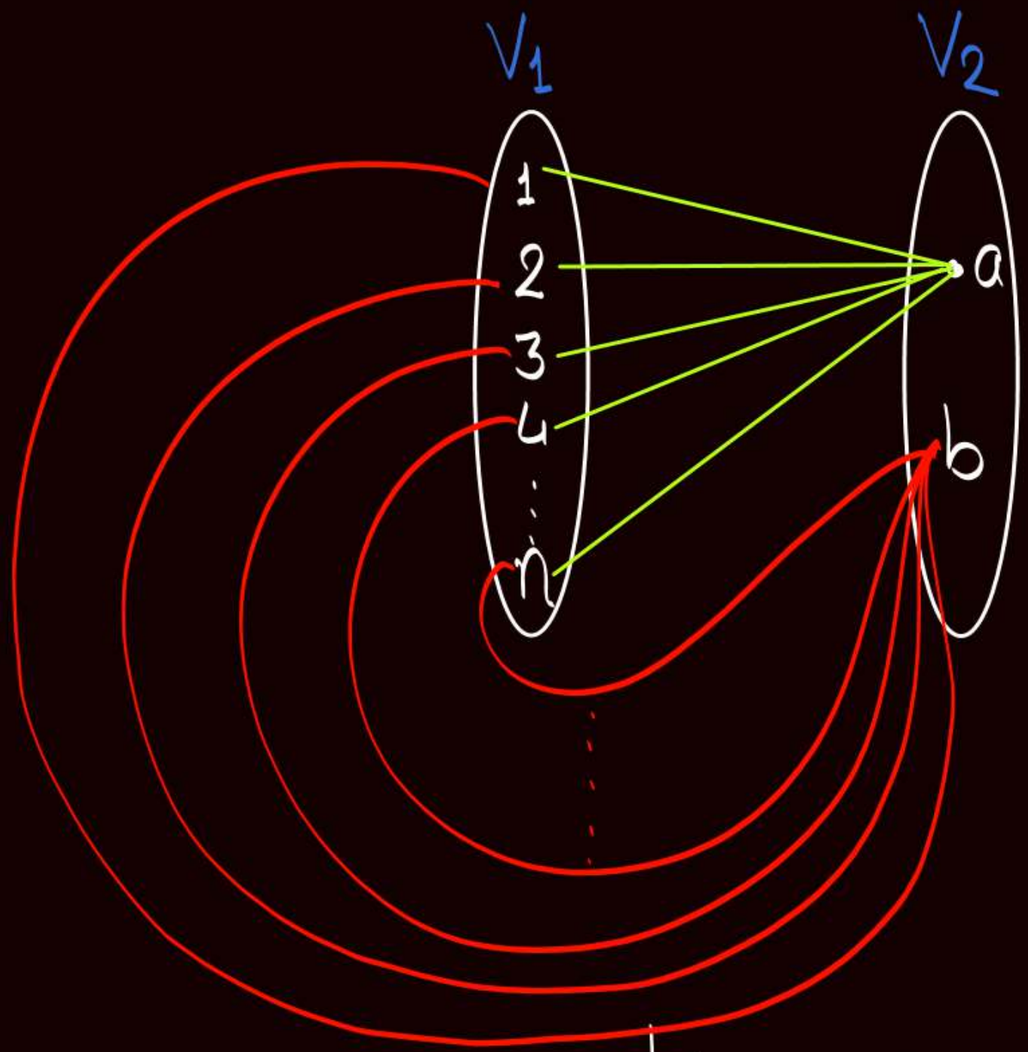
$\therefore K_{1,n}$ is also planar



$K_{2,2}$ is planar



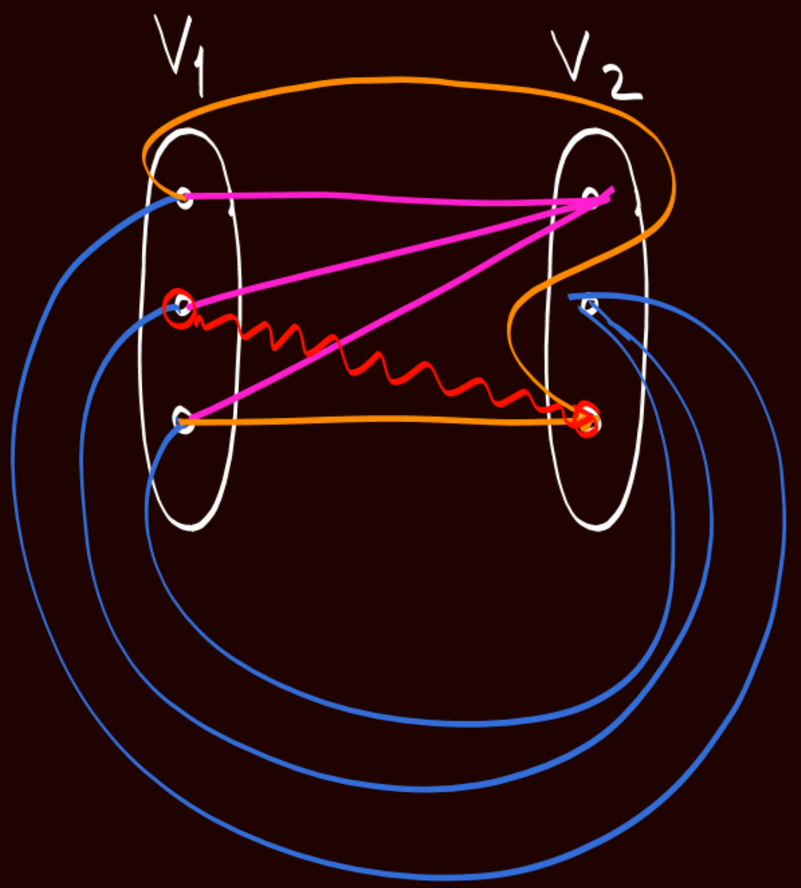
$K_{3,2}$ is planar



$K_{n,2}$ is planar

$\therefore K_{2,n}$ is also planar

* Check whether $K_{3,3}$ is a planar graph or not?



$K_{3,3}$ is non-planar

Complete bipartite graph $K_{m,n}$ is planar if and only if $m \leq 2$ or $n \leq 2$

$K_{3,3}$ is a non-planar graph with '6' vertices but with only '9' edges

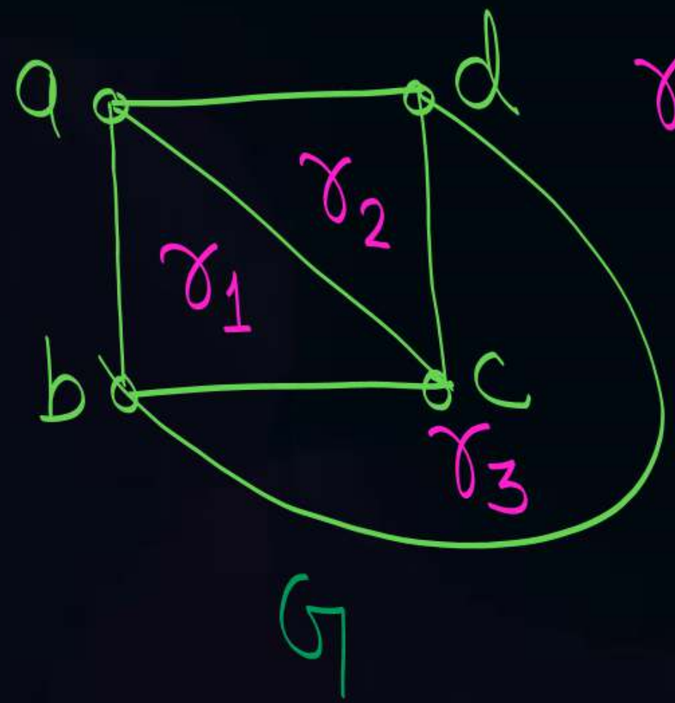
Note:-

- (1) K_5 is the non-planar graph with minimum number of vertices.
  it has only 5 vertices but 10 edges
- (2) $K_{3,3}$ is a non-planar graph with minimum number of edges.
  it has only 9 edges, but 6 vertices



Topic : Region / Face

- Every planar graph divides a plane into connected areas called as regions of the plane or faces of the planar graph.



- γ_1, γ_2 and γ_3 are bouded regions (interior regions) of planar graph G

Whereas γ_4 is an unbounded (Exterior) region of Planar graph G.



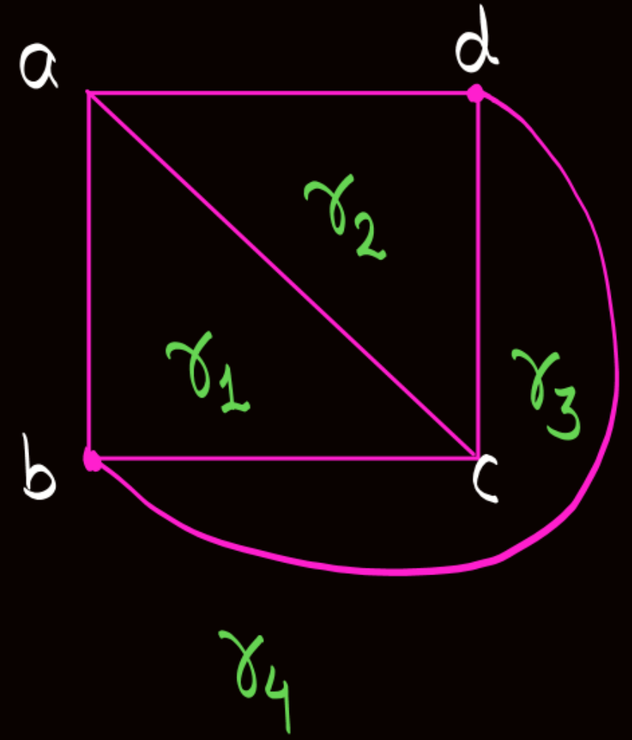
Topic : Region / Face

- In any planar graph there will be exactly '1' Unbounded (Exterior) region, all other regions of the Planar graph will be bounded (interior) regions.

- Number of bounded regions (faces) in a planar graph = Total no. of regions in that Planar graph − 1

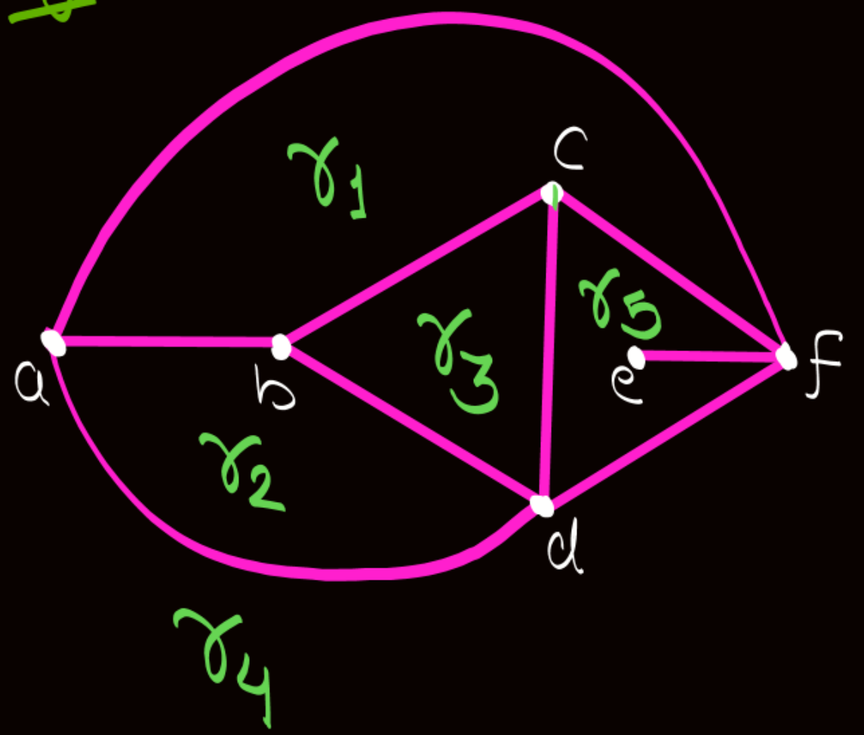
eg 1:

G_1


a
d
b
c
γ_1
γ_2
γ_3
γ_4

eg 2:-

G_2


γ_1
c
γ_3
γ_5
a
b
e
f
γ_2
d
γ_4



Topic : Degree of a region/face

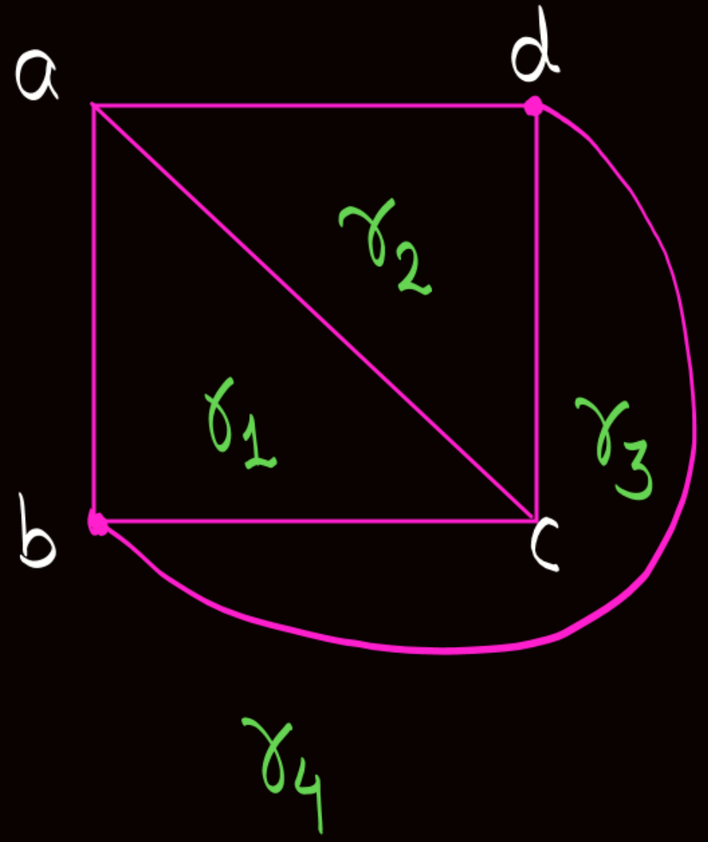
* Degree of a region "γ" can be denoted by $\deg(\gamma)$, and it is defined as.

$\deg(\gamma)$ = No. of edges Enclosing the region 'γ'

(or)

No. of Edges Exposed to region 'γ'

eg:-



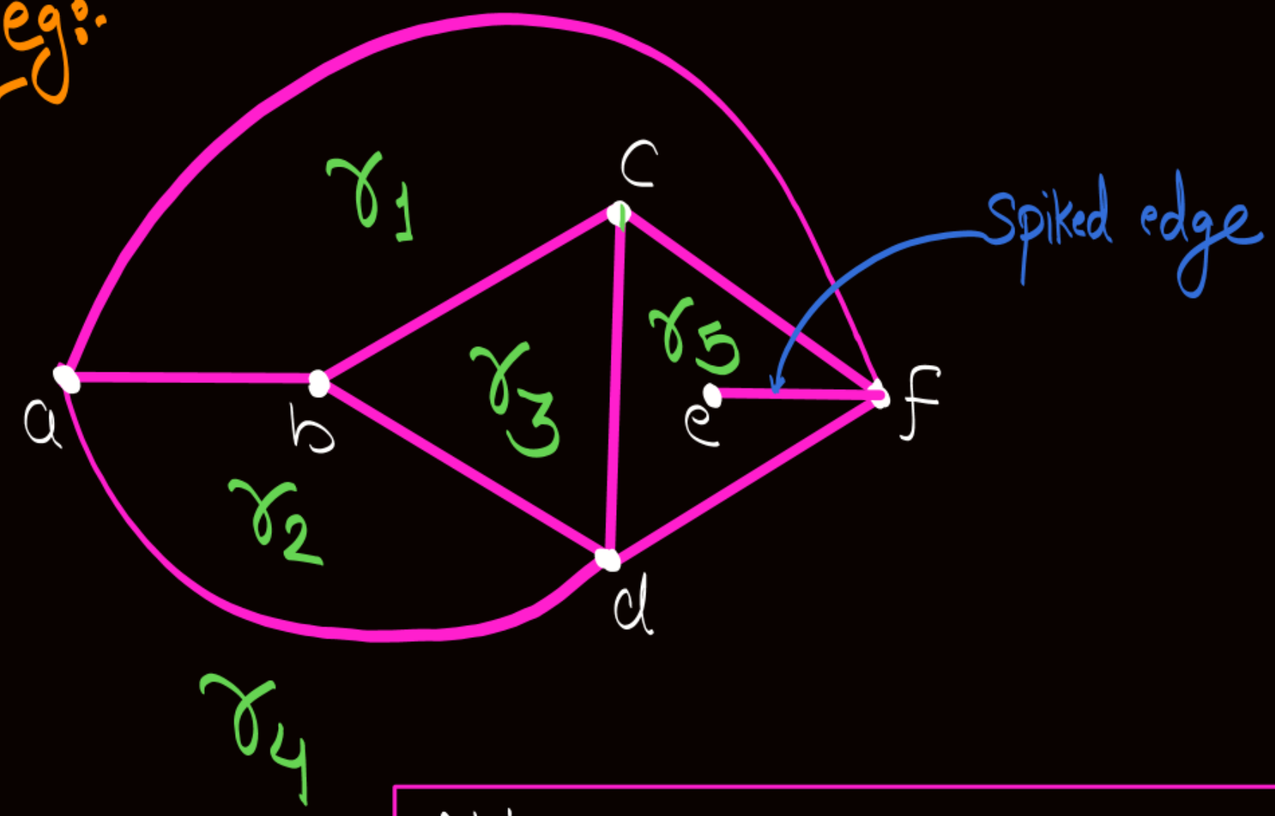
$\deg(r_1) = 3$

$\deg(r_2) = 3$

$\deg(r_3) = 3$

$\deg(r_4) = 3$

eg:-



$\deg(r_1) = 4$

$\deg(r_2) = 3$

$\deg(r_3) = 3$

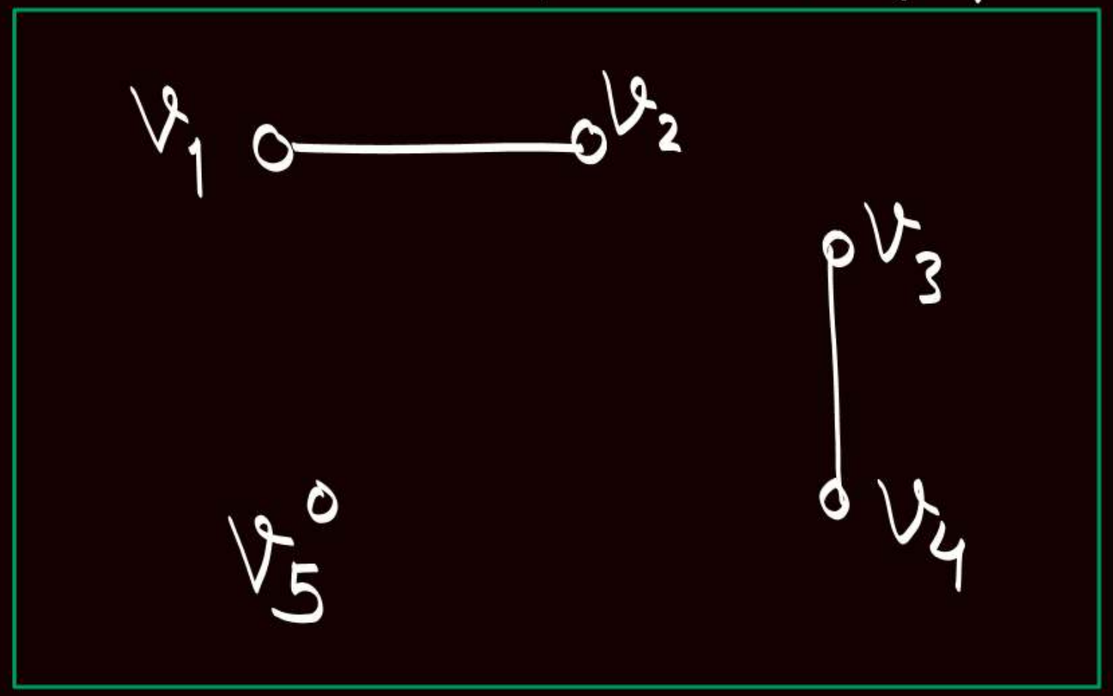
$\deg(r_4) = 3$

$\deg(r_5) = 3+2 = 5$

because of Spiked Edge

Note:- Spiked Edge will be Counted as two Edges to obtain the degree of Corresponding region

Consider the following graph G



← it is also a planar graph with only '1' region

Lets call it region 'r'

then $\deg(r) = 2 + 2 = 4$

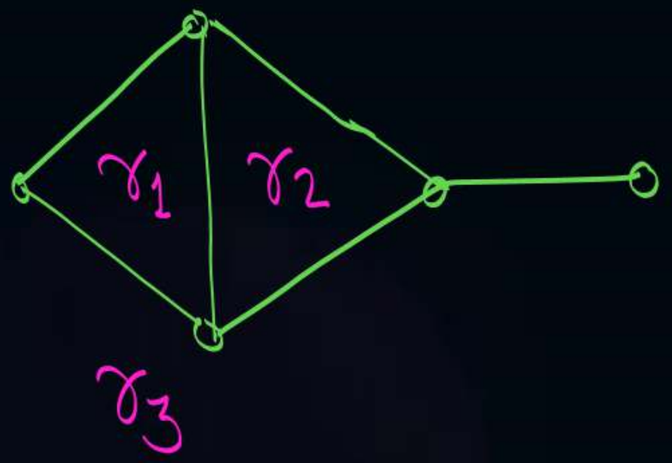
No two edges cross each other at a non-vertex point,

∴ graph G is a planar graph



Topic : Sum of degree theorem

w.r.t. degree of regions



* Let $|R|$ denotes the number of regions in the planar graph $G = (V, E)$. then

$$\sum_{i=1}^{|R|} \deg(r_i) = 2|E|$$

$$\deg(r_1) + \deg(r_2) + \deg(r_3) = 2 * |E|$$

$$3 + 3 + 6 = 2 * 6$$

$$= 12 \qquad = 12$$

Some authors denotes the no. of regions by $|R|$, and some author denote the no. of faces by 'f' and $|R| = f$



Topic : Sum of degree theorem

- **Corollary 1:-** No. of regions of odd degree are always even

- **Corollary 2:-** In a planar graph G, if degree of each region is Exactly 'k' then

$$k.|R| = 2|E|$$



Topic : Sum of degree theorem

Corollary 3: In a planar graph G if degree of each region is at least 'K' {i.e. $\geq K$} then

$$K|R| \leq 2|E|$$



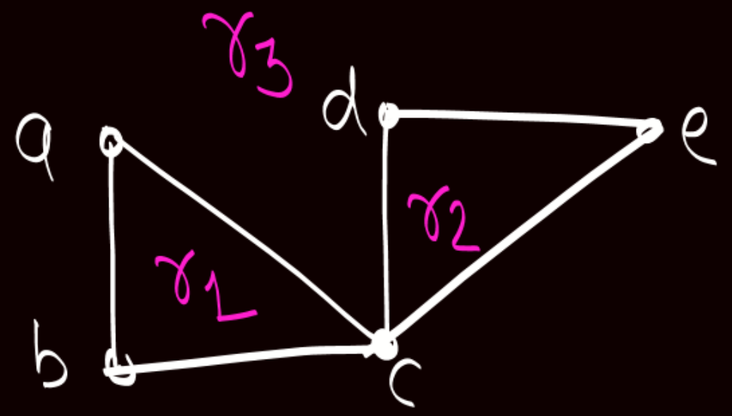
Topic : Sum of degree theorem

Corollary 4: In a planar graph G if degree of each region is at most 'k' {i.e $\leq k$} then

$$k|R| \geq 2|E|$$

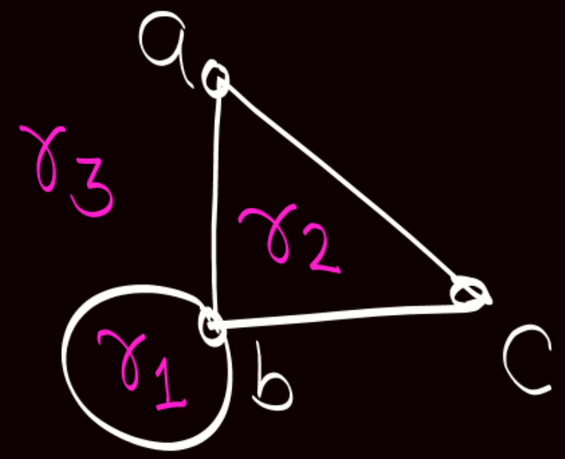
Note:- Every planar graph need not be simple,

eg.



Simple planar graph

(No loop
+
No parallel edges)



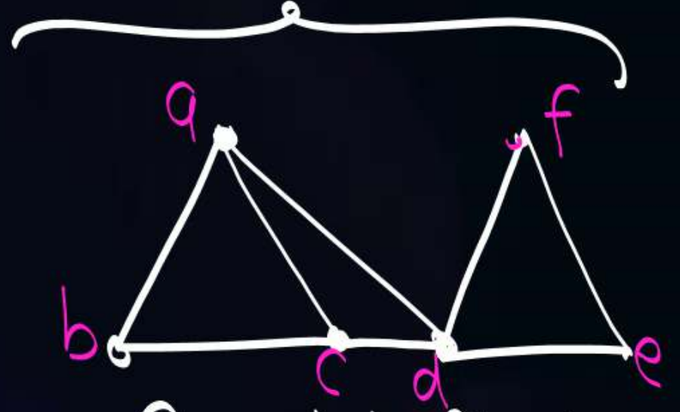
It is also a planar graph, but it is not simple graph



Topic : Simple planar graph

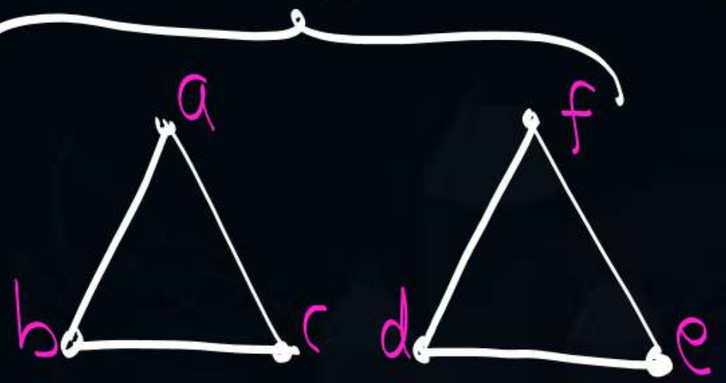
* A planar graph with no loop and no parallel edges is called a simple planar graph { It may or may not be Connected. }

Eg. G_1



Connected Simple Planar graph

Eg. G_2



It is a simple planar graph, but not a connected graph

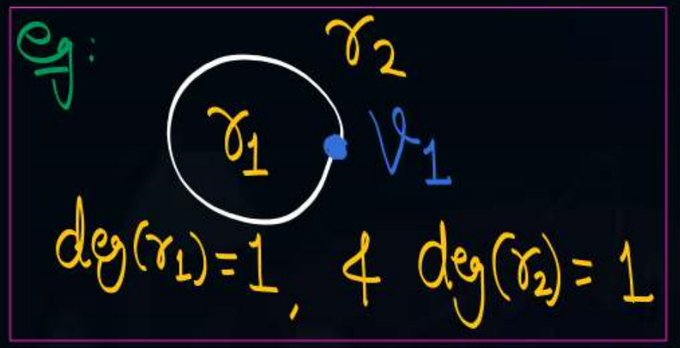


Topic : Simple planar graph

Note:- In a planar graph with two or more regions.
{i.e, in a planar graph with at least two regions}

(i) $\deg(r) = 1$, if and only if there is a loop

eg:



$\deg(r_1) = 1$, & $\deg(r_2) = 1$

& (ii) $\deg(r) = 2$, if and only if region is w.r.t. parallel edges

eg:



$\deg(r_1) = 2$
& $\deg(r_2) = 2$



Topic : Simple planar graph

- In a simple planar graph with two or more regions degree of each region will be at least '3' { it may or may not be Connected }

In a planar graph if degree of each region is at least '3', then

$$3|R| \leq 2|E|$$

will hold true

$\Rightarrow$ ∴ In a simple planar graph with two or more regions

$$3|R| \leq 2|E|$$

will always hold true



Topic : Simple planar graph

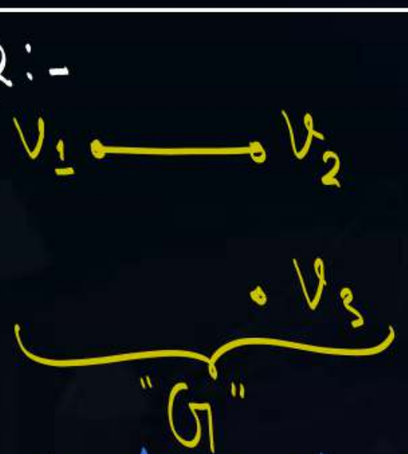
- In a simple planar graph with two or more regions degree of each region will be at least '3'

eg 1:-



It is a simple Planar graph with only '1' region and deg(region) = 0

eg 2:-



It is also a simple Planar graph with only '1' region and deg(region) = 2



Topic : Euler's equation (For Connected Planar Graph)

- For any connected planar graph G, {may or may not be a simple graph} following Equation will hold true

$$|V| + |R| = |E| + 2$$

← It is called Euler's Equation for Connected planar graph

Where $|V|$ = No. of vertices
$|E|$ = No. of edges
$|R|$ = No. of regions

$|V| + |R| = |E| + 2$ will not hold true if planar graph is not a Connected graph

Some author represent the Euler's Equation using the following Format.

$$n + f - e = 2$$

Where,
n = No. of vertices
e = No. of edges
f = No. of faces = No. of regions

Note 1:- In a simple planar graph {may or may not be Connected}

$$3|R| \leq 2|E| \quad \text{---- eq}^{n} \text{ (1)}$$

will always hold true

Note 2: In a connected planar graph {may or may not be simple}

$$|V| + |R| = |E| + 2 \quad \text{---- eq}^{n} \text{ (2)}$$

will always hold true



Topic : Simple connected planar graph

Simple Connected planar graph $\Rightarrow$ Simple planar graph + Connected Planar graph

Simple planar graph $\rightarrow$ ($\because 3|R| \leq 2|E|$ will hold true)

and

Connected Planar graph $\rightarrow$ ($\because |V|+|R| = |E|+2$ will also hold true)

① In a simple planar graph only eqⁿ ① is guaranteed to hold true, but eqⁿ ② may or may not hold true

② In a Connected Planar graph only eqⁿ ② is guaranteed to hold true, but eqⁿ ① may or may not hold true

③ If graph is a simple Connected Planar graph, then both eqⁿ ① & eqⁿ ② will hold true



Topic : Simple connected planar graph

Simple Connected planar graph $\Rightarrow$ Simple planar graph + Connected Planar graph

Simple planar graph $\Rightarrow$ $\therefore 3|R| \leq 2|E|$ - eq^n (1)

Connected Planar graph $\Rightarrow$ $\therefore |V|+|R| = |E|+2$ - eq^n (2)

From Equation (1) $3|R| \leq 2|E| \therefore |R| \leq \frac{2|E|}{3}$

Put this value of $|R|$ in eq^n (2)

$$|V| + \frac{2|E|}{3} \geq |E|+2$$

$$|V| \geq \frac{|E|}{3} + 2 \Rightarrow \boxed{|E| \leq 3|V|-6} \text{ - eq}^n \text{ (3)}$$

(or) $\boxed{e \leq 3n-6}$

From eq^n (1) $3|R| \leq 2|E| \therefore |E| \geq \frac{3|R|}{2}$

Put this value of $|E|$ in eq^n (2)

$$|V| + |R| \geq \frac{3|R|}{2} + 2$$

$$|V| \geq \frac{|R|}{2} + 2 \Rightarrow \boxed{|R| \leq 2|V|-4} \text{ - eq}^n \text{ (4)}$$

$\boxed{n \geq \frac{f}{2} + 2}$ or $f \leq 2n-4$



Topic : Note

In a simple Connected planar graph all four equations will always hold true

$$3|R| \leq 2|E| \quad \text{——} (1)$$

$$|V|+|R| = |E|+2 \quad \text{——} (2)$$

$$|E| \leq 3|V|-6 \quad \text{——} (3)$$

$$|R| \leq 2|V|-4 \quad \text{——} (4)$$



Topic : Polyhedral graph

* A Polyhedral graph is a Simple Connected planar graph in which degree of each 'Vertex' is at least '3'.

→ $\therefore 3|V| \leq 2|E|$ —— eqn (5)

will also hold true

- In a polyhedral graph, following equations will hold true

$$\begin{cases} 3|R| \le 2|E| & \text{---(1)} \\ |V|+|R| = |E|+2 & \text{---(2)} \\ |E| \le 3|V|-6 & \text{---(3)} \\ |R| \le 2|V|-4 & \text{---(4)} \\ 3|V| \le 2|E| & \text{---(5)} \end{cases}$$

From eqⁿ (5) $3|V| \le 2|E| \quad \therefore |V| \le \frac{2|E|}{3}$

Put this value of $|V|$ in Euler's eqⁿ

$$\frac{2|E|}{3} + |R| \geqslant |E|+2$$

$$|R| \geqslant \frac{|E|}{3}+2 \Rightarrow \boxed{|E| \le 3|R|-6} \quad \text{eqⁿ (6)}$$

From eqⁿ (5) $3|V| \le 2|E| \quad \therefore |E| \geqslant \frac{3|V|}{2}$

Put this value $|E|$ in Euler's equation

$$|V|+|R| \geqslant \frac{3|V|}{2}+2$$

$$|R| \geqslant \frac{|V|}{2}+2 \Rightarrow \boxed{|V| \le 2|R|-4} \quad \text{eqⁿ (7)}$$

Note: In a polyhedral graph all seven equations from eqn (1) to eqn (7) will hold true

H.W. :- Obtain Euler's Equation for disconnected Planar graph with 'k' Connected Components in it.

∴ Euler's equation for a disconnected planar graph with 'k' Connected Components



2 mins Summary

Topic Planar graphs

Slide

THANK - YOU